SEVO: Semantic-Enhanced Virtual Observation

System Pipeline & Parameter-Level Architecture Analysis

STANDARD BASELINE Raw RGB → Policy → Action

✕ **Fails in novel environments**

⚠️ Most community setups use wrist cameras

SEVO PIPELINE (Ours)

— unified observation design, applied identically during collection & deployment

BODY-FIXED CAMERAS

Static relative to robot body

Combined FOV covers full task

Front | Side | TOP

Standard RGB, no depth

Wrist = control only ✕

SEVO ENHANCEMENT

YOLO Segmentation

YOLOv8-seg @ ~30 FPS

Binary mask $M_t \in \{0,1\}^{H \times W}$

Category filter: "bottle"

Red Illumination

5W LED (620–630 nm)

Physical domain normalizer

True 3D geometry anchor

synergy

Per-Pixel Mask Overlay

$\tilde{I}_t = (1 - \alpha \cdot M_t) \odot I_t + \alpha \cdot M_t \odot C$

$\alpha = 0.45$ $C = [255,255,0]$

Yellow highlight on target

→ **background-invariant $\tilde{I}_t$**

Pixel (r,g,b) → (0.55r+115, 0.55g+115, 0.55b)

VIRTUAL CAMERA

Enhanced $\tilde{I}_t$

+ Joint state $q_t \in \mathbb{R}^6$

+ Language (SmolVLA)

Stable across all environments

POLICY π

ACT (80 episodes)

SmolVLA (120 ep.)

Unmodified arch.

Action

$a_t \in \mathbb{R}^6$

✓ 95% ACT 83% SmolVLA

Cross-environment success (body-fixed cameras)

DATA COLLECTION PROTOCOL

— #1 dominant factor for generalization

Varied Backgrounds

#1 — forces feature disentanglement

Varied Lighting

On / off / dark + red LED on

Distractors + Nulls

Clutter + empty-scene episodes

YOLO–Motion Binding

Arm moves only when overlay visible

Component Importance Ranking:

Varied BG > Red Light > YOLO Overlay